Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Validation of a Cephalad Fluid Shift Countermeasure		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	brandon.r.macias@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	06/01/2015	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Change in period of performance per NSBRI (formerly 5/1/15-4/30/17)--Ed., 7/7/15		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sargsyan, Ashot M.D. (Wyle Laboratories) Cole, Christine (Clemson University) Kesari, Santosh M.D., Ph.D. (John Wayne Cancer Institute) Stenger, Michael Ph.D. (Wyle Laboratories) Ebert, Douglas Ph.D. (Wyle Laboratories) Hargens, Alan Ph.D. (University of California, San Diego) Liu, John Ph.D. (University of California, San Diego) Lee, Stuart M.S. (Wyle Integrated Sciences and Engineering Group) Huang, Alex M.D., Ph.D. (University of California, Los Angeles)		
Grant/Contract No.:	NCC 9-58-SMST04202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	RATIONALE: In humans exposed to up to ~6 months of weightlessness, altered visual acuity and structural adaptions of the ocular system have developed, including hyperopic shifts, choroidal folds, and optic disc edema. While the etiology and progression of vision impairment and intracranial pressure (VIIP)/Spaceflight Associated Neuro-ocular Syndrome (SANS) are not yet clearly defined, given the potential implications for mission success and long-term health, potential countermeasures are being investigated in parallel with definition of VIIP/SANS mechanisms. To prepare for future exploration missions, efforts currently are directed to identify factors that contribute to the ocular structural and functional changes in parallel with evaluating possible countermeasures using ground-based analogs of weightlessness. Given that the headward fluid shift associated with weightlessness has been hypothesized to be one of the central contributors to VIIP/SANS, countermeasures that reverse the headward fluid shift, such as those that trap fluid in the legs or lower body, may have the greatest promise. The goal of this project was to determine if venoconstrictive thigh cuffs used during acute head-down tilt (HDT), a spaceflight analog, mitigate changes in intracranial pressure (ICP), ocular, and intraocular pressure (IOP) associated with a cephalad fluid shift. STRATEGY & STATUS: This project's overall strategy was to develop an enhanced thigh cuff that can be comfortably worn for extended periods during spaceflight and conduct human experiments to determine if a thigh cuff countermeasure mitigates a HDT-induced cephalad fluid shift. The study was designed to follow a stepwise process and was completed in two years. In Year 1, our team tested the efficacy of commercially-available thigh cuffs in healthy normal subjects and developed a prototype enhanced cuff. Dr. Macias' team in collaboration with co-investigator, Dr. Huang of Doheny-UCLA, enrolled 20 healthy normal subjects to evaluate pneumatically-controlled thigh cuffs during head-down tilt. From this Year-1 work, Drs. Macias and Huang presented two abstracts at The Association for Research in Vision and Ophthalmology 2017 Annual Meeting and wrote a manuscript for publication in a peer-reviewed journal. The primary findings from this work suggest that thigh cuffs mitigate head-down-tilt induced increases of intraocular pressure and choroid thickness. Concurrently, Dr. Cole and her team at Clemson University designed and produced novel, prototype thigh cuffs to address previously identified problems with the thigh cuffs currently used in the space, the Russian Braslet. For example, some of these problems include several layers of material to generate elastic force, bunching of material at the buckle, an inability to fine tune cuff compression. Functional enhancements included usage of lighter weight, more flexible materials in fewer layers; improved closures; and ability to focus compression on the thigh (femoral vein). Dr. Cole focused on achieving enhanced cuff performance by using well-characterized materials that are commercially available. During Year 2, Dr. Cole's team completed fabrication of three sets of enhanced thigh cuffs (final design) in four different sizes to accommodate a variety of thigh circumferences and delivered them to the investigator team to support the human testing. Dr. Huang enrolled 20 additional healthy normal subjects to evaluate these enhanced cuffs using the same test protocol as used in the evaluation of the standard thigh cuffs in Year 1. Drs. Macias and Kesari also completed tests of three patients from the John Wayne Cancer Institute in whom direct ICP measurements were possible. Measurements were obtained with and without the enhanced thigh cuffs. The primary findings from this work suggest that the enhanced thigh cuff mitigates head-down-tilt induced increases in intraocular pressure, and choroid thickness. However, the thigh cuff device did not significantly alter optic nerve sheath diameter or noninvasive measures of intracranial pressure. Additionally, a UCSD (University of California San Diego) Bioengineering Senior Design Team supervised by Dr. Hargens evaluated skin contact pressure provided by the enhanced thigh cuff. Further, Dr. Hargens' UCSD research group also identified components that could be made in the development of a combined lower body negative pressure (LBNP) and thigh cuff countermeasure suit. In the course of this two-year project, the investigators demonstrated the efficacy of thigh cuffs to prevent some ocular and hemodynamic changes associated with a fluid shift induced by acute head-down tilt, designed and constructed enhanced thigh cuffs designed to improve fit, comfort, and performance in comparison to the Russian Braslet, and demonstrated the efficacy of these enhanced cuffs in healthy normal subjects and in patients.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Currently, there are few strategies available to treat intracranial hypertension, intracranial pressure, and pressure related vision disease. This project demonstrates the ability for a thigh cuff device to attenuate a head-down-tilt induced elevation in intraocular pressure and choroid thickness. Therefore, this countermeasure may provide clinicians with a new tool to alter intraocular pressure.
Task Progress:	STRATEGY & STATUS: This project's overall strategy was to develop an enhanced thigh cuff that can be comfortably worn for extended periods during spaceflight and conduct human experiments to determine if a thigh cuff countermeasure mitigates a HDT-induced cephalad fluid shift. The study was designed to follow a stepwise process and was completed in two years. In Year 1, our team tested the efficacy of commercially-available thigh cuffs in healthy normal subjects and developed a prototype enhanced cuff. Dr. Macias' team in collaboration with co-investigator Dr. Huang of Doheny-UCLA, enrolled 20 healthy normal subjects to evaluate pneumatically-controlled thigh cuffs during head-down tilt. Concurrently, Dr. Cole and her team at Clemson University designed and produced novel, prototype thigh cuffs to address previously identified problems with the thigh cuffs currently used in the space, the Russian Braslet. For example, some of these problems include several layers of material to generate elastic force, bunching of material at the buckle, an inability to fine tune cuff compression. Functional enhancements included usage of lighter weight, more flexible materials in fewer layers; improved closures; and ability to focus compression on the thigh (femoral vein). Dr. Cole focused on achieving enhanced cuff performance by using well-characterized materials that are commercially available. During Year 2, Dr. Cole's team completed fabrication of three sets of enhanced thigh cuffs (final design) in four different sizes to accommodate a variety of thigh circumferences and delivered them to the investigator team to support the human testing. Dr. Huang enrolled 20 additional healthy normal subjects to evaluate these enhanced cuffs using the same test protocol as used in the evaluation of the standard thigh cuffs in Year 1. Drs. Macias and Kesari also completed tests of three patients from the John Wayne Cancer Institute in whom direct ICP measurements were possible. Measurements were obtained with and without the enhanced thigh cuffs. Further, Dr. Hargens' UCSD research group also identified components that could be made in the development of a combined lower body negative pressure (LBNP) and thigh cuff countermeasure suit.

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Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Challa ST, Hargens AR, Uzosike A, Macias BR. "Muscle microvascular blood flow, oxygenation, pH, and perfusion pressure decrease in simulated acute compartment syndrome." J Bone Joint Surg Am. 2017 Sep 6;99(17):1453-9. https://doi.org/10.2106/JBJS.16.01191 ; PubMed PMID: 28872527; PubMed Central PMCID: PMC5685422 , Sep-2017
Articles in Peer-reviewed Journals	Christian KH, Petitti C, Ortega-Schwartz K, Mulder E, Noppe A, von der Wiesche M, Stern C, Young M, Macias BR, Laurie SS, Lovering AT. "Development of optic disc edema during 30 days of hypercapnic head-down tilt bedrest is associated with short sleep duration and blunted temperature amplitude." J Appl Physiol (1985). 2024 Apr 1;136(4):753-763. https://doi.org/10.1152/japplphysiol.00211.2023 ; PMID: 38357726 ; PMCID: PMC11286268 , Apr-2024
Articles in Peer-reviewed Journals	Huang AS, Jalili J, Walker E, Weinreb RN, Laurie SS, Macias BR, Christopher M. "Artificial intelligence deep learning models to predict spaceflight associated neuro-ocular syndrome (SANS)." Am J Ophthalmol. 2025 Oct;278:115-123. https://doi.org/10.1016/j.ajo.2025.06.009 ; PubMed PMID: 40505849; PubMed Central PMCID: PMC12481839 , Oct-2025
Articles in Peer-reviewed Journals	Svoronos AA, O'Grady CS, Walker E, Afshari NA, Macias BR, Laurie SS, Weinreb RN, Huang AS. "Analysis of spaceflight-associated biometric and refractive changes in astronauts." Am J Ophthalmol. 2025 Aug;276:146-56. http://dx.doi.org/10.1016/j.ajo.2025.04.001 ; PMID: 40194644 , Aug-2025
Articles in Peer-reviewed Journals	He Y, Karanjia R, Zhang X, Wanderer D, Walker E, Lee SH, Poczatek MJ, O'Grady CS, Marion KM, Young M, Moestl S, Mulder E, Stern C, Sadda SR, Laurie SS, Macias BR, Huang AS. "Optic nerve vasculature and countermeasure assessment in a bedrest analogue of spaceflight-associated neuro-ocular syndrome." Am J Ophthalmol. 2025 Oct;278:317-327. https://doi.org/10.1016/j.ajo.2025.06.032 ; PubMed PMID: 40545016; PubMed Central PMCID: PMC12462775 , Oct-2025
Articles in Peer-reviewed Journals	Laurie SS, Lee SMC, Macias BR, Patel N, Stern C, Young M, Stenger MB. "Optic disc edema and choroidal engorgement in astronauts during spaceflight and individuals exposed to bed rest." JAMA Ophthalmol. 2019 Dec 26;138(2):165-72. https://doi.org/10.1001/jamaophthalmol.2019.5261 ; PMID: 31876939; PMCID: PMC6990717 , Dec-2019
Articles in Peer-reviewed Journals	Balasubramanian S, Tepelus T, Stenger MB, Lee SMC, Laurie SS, Liu JHK, Feiveson AH, Sadda SR, Huang AS, Macias BR. "Thigh cuffs as a countermeasure for ocular changes in simulated weightlessness." Ophthalmology. 2018 Mar;125(3):459-60. Epub 2017 Nov 16. https://doi.org/10.1016/j.ophtha.2017.10.023 ; PubMed PMID: 29153458 , Mar-2018
Awards	Hargens A. "NASA Distinguished Public Service Medal, June 2017." Jun-2017
Awards	Macias B. "KBRwyle Excellence in Science Award, July 2017." Jul-2017